Enrolment Number										

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Winter, 2026

B.Tech 4th Semester Examination

Transportation Engineering – Theory

Course Code: BCE23401T

Total Marks – 50 **Time – 2 hours**

SECTION 1

Answer all the questions **(Multiple Choice) Marks: 10 x 1 =10**

I.

I. The major purpose of reconnaissance survey in highway planning is to:
a) Finalize pavement thickness
b) Collect detailed soil test data
c) Identify feasible alternative routes
d) Prepare final cost estimates

II. Aggregate Crushing Value (ACV) indicates the aggregate's resistance to:
a) Abrasion
b) Weathering
c) Crushing under compressive load
d) Water absorption

III. A single-lane pavement designed as per IRC standards should have a carriageway width of:
a) 3.5 m
b) 3.0 m
c) 3.75 m
d) 7.0 m

IV. In a CBR test, if the load sustained by the soil sample at 2.5 mm penetration is 820 kg, then the CBR value is approximately:
a) 40%
b) 50%
c) 60%
d) 70%

V. The coefficient of lateral friction required on a curve mainly depends upon:
a) Traffic volume only
b) Speed and radius of curve
c) Pavement colour
d) Shoulder width

VI. What percentage camber must be provided for a CC road passing through low rainfall area?
a) 3.0 %
b) 2.5 %

c) 2.0 %
d) 1.7 %

VII. Full form of CRF is
a) Central Rail Fund
b) Central Road Finance
c) Central Road Fund
d) Central Road Facility

VIII. What is the minimum design speed on a National Highway in a plain terrain?
a) 70 kmph
b) 60 kmph
c) 100 kmph
d) 80 kmph

IX. If a vehicle travels at 90 km/h on a curve of radius 400 m, the theoretical super elevation required is closest to:
a) 0.05
b) 0.09
c) 0.15
d) 0.20

X. Bitumen is a by-product of _____
a) Wood
b) Petroleum
c) Kerosene
d) Coal

SECTION 2

2. Answer any one question from this section **Marks: 8**

A) Describe the design of walkways and tactile paving systems for visually impaired users. Include dimensions, materials, and practical considerations.

B) Explain the desirable properties of aggregates used in highway construction. Discuss how aggregate crushing and abrasion tests help in evaluating pavement materials.

SECTION 3

3. Answer any one question from this section **Marks: 8**

A) Discuss the importance of traffic engineering studies in modern transportation systems. Explain how traffic volume, speed, and accident studies help in traffic management.

B) Compare flexible pavements and rigid pavements with respect to design, load transfer mechanism, materials used, and maintenance requirements.

SECTION 4

4. Answer any one question from this section **Marks: 8**

A) Derive the expression for Overtaking Sight Distance for a road having two-way traffic.

B) Discuss the factors affecting sight distances. Also derive the expression for Stopping Sight Distance.

SECTION 5

5. Answer any one question from this section **Marks: 8 (4+4)**

A)

1. What is extra widening of highway? Why is it necessary?
2. Calculate the extra widening required for a NH on a horizontal curve of radius 270m for a design speed of 70 kmph. Assume the longest wheel base to be travelled on the road as 7 m.

B)

1. Why is the purpose of using camber on a highway? Explain with illustration.
2. In Meghalaya, an MDR of WBM pavement 4 m wide and another State Highway of bituminous concrete pavement 7 m wide are to be constructed. What should be the height of the crown w.r.t. the edges in the mentioned two cases?

SECTION 6

6. Answer any one question from this section **Marks: 8**

A) Write A national highway is passing through a rolling terrain of heavy rainfall area where the radius of curve is 550 m. Design the length of the transition curve required for the section.

B) A CBR test was conducted over soil sub grade and following results were obtained. Assume curve throughout convex following materials are required to be used over the soil sub grade.

i. Compact soil (CBR= 6%)
ii. Poorly graded gravel (CBR= 12%)
iii. Well graded gravel (CBR= 60%)
iv. Thickness of bituminous surface = 3.85 cm

Design the pavement and find the thickness of the bituminous surface layer placed over the well graded gravel using CBR method if the wheel load is 4100 kg and tyre pressure is 7 kg/cm^2.

Penetration (mm)	2.5	5
Load (kg)	60	82

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